

Paper Directions 23 — my read (Aug 18, 2026)

Source: `transcripts/20260918_Paper_directions_23_otter_ai_transcript.docx` (~40 min)
Context: Charles walked Alex through the PD20–22 memo / HAND backup slides (box QC, minutes floor, ρ bracket, ppm0lt20 contrast). First live read-aloud of the full campaign arc.
Prior: `PD21_notes.md` · `PD20_notes.md` · `../3-Master_Plan/re_entry/HEROs_and_PASSes/PD20_22_campaign_big_picture.md`

Headline

Alex is **thrilled with the hero and the panel story** — defensive PD22 work did not break the main result. **Next mission is MLE:** write the likelihood, then fit λ , γ , t^* on fixed empirical rosters. ρ bracket / H_{sort} forensics and the ppm-zero std-dev plot are **parked** ($\leq$ one day) — not blocking the paper.

What Charles covered (briefing arc)

Charles walked the full ladder: Army inverted-U $\rightarrow$ MBB hero $\rightarrow$ Wang-style minimal model (congestion in score) $\rightarrow$ LG pipeline (ASSIGN $\rightarrow$ SCORE $\rightarrow$ SELECT) $\rightarrow$ Gibbs gate cleared (PD20) $\rightarrow$ ρ calibration via H_{sort} bracket $\rightarrow$ box QC + minutes policy (PD22) $\rightarrow$ ppm0lt20 contrast slides.
Alex said the articulation practice was valuable; for his book the scientific state is unchanged from “a little while ago” except that panel assumptions are now defended.

Alex reactions — what landed

Topic	Alex
Box QC (dash rows, ≥ 11 games)	“This looks great.”
Roster-size before/after	Happy — tails gone
ESPN 2013 $\rightarrow$ 2014 depth jump	OK — Charles has a finger on it
Drafted-player minutes CDF	Misread once (44 = drafted player-seasons lost at min 20, not total players); cleared
$\rho^* = 0$ on hero panel	Not worried — modest H_{sort} fit, not “NCAA is random”
Interval overlap / visual sorting	Accepts geometry vs bracket ρ^* are different questions
ppm0lt20 ρ^* “strong”	“Where’s the problem?” — if hero + intervals + Gibbs still work, fine
$\rho \approx 0$ as must-have	No — “edge calculation”; hero inverted-U is the main result
Drop 2011–2012 seasons	Yes — treat as data-quality; analysis from 2013+ is defensible
End state	Same as before defensive push: fit λ , γ , t by MLE

ρ , H_{sort} , ppm-zero — clarify for next briefing

Hero panel (slide 14, locked): drop sub-20 + box QC + roster caps $\rightarrow$ bracket **$\rho^* = 0$** all seasons, $H_{\text{sort}} \approx 0.06$. Model–measurement fit, not “no sorting in NCAA.”
ppm0lt20 contrast (slide 15): real Aug 17 bracket on **that estimand** $\rightarrow$ longitudinal $\rho^* \approx 0.57$, per-season ρ^* non-zero. Alex liked this as evidence the model **can** track empirical sorting when targets are higher.
Important nuance (post-meeting): the ppm0 bracket JSON was fit when the ppm-zero panel still had **pre–box-QC ingestion** (~10k player-seasons/season in 2014 vs ~5.6k after QC). Charles is **re-running H_{sort} calibration** on the current pipeline. After box QC, empirical H_{sort} under ppm-zero $\approx$ drop ($\Delta \approx 0.001$) — so contrast slide 15 shows **what ppm0lt20 implied on the old ingestion panel**, not necessarily what a fresh bracket on today’s ppm-zero panel would give.
Locked production policy (do not confuse with Otter end): drop sub-20, not ppm-zero. Otter garbled the closing (“you’re out” / “it’s zero”). Alex’s operational lock at end of call mixed drop + ppm-zero + drop 2011–2012 — **write down explicitly:** drop-at-20 for ASSIGN input; ppm-zero stays **contrast only**; consider **2013–2021** as primary window.

“Where’s the problem?” — Charles vs Alex

Charles was still carrying PD22 worry: all-zero ρ^* on drop panel felt scary vs interval-overlap pictures.
Alex’s frame:

- Hero inverted-U + high visual sorting + Gibbs survival = **main line intact**
- ρ^* small or zero after cleaning assumptions is **not fatal**
- Revisiting cleaning assumptions was the **point** of PD22 — no blame for moving fast earlier
- If hero survives, **move on to MLE**

Charles corrected Alex once: **ρ is not a dial that turns up congestion** — congestion needs viable competition (γ -sharp peers); weak leagues don’t get “more congestion” from ρ alone.

Std-dev vs zero-fraction plot (slide 18 area)

Alex pushed on the **positive slope**: more ppm-zero'd players per team → **higher** within-team perf std — opposite naive “everyone at zero → lower dispersion → higher homophily.”

Alex intuition: identical zeros should **lower** team dispersion → **raise** H_sort, not the direction shown.

Alex priority: worth ≤ **one day** — inspect H_sort **numerators** (team-level SS decomposition), verify perf column uses z-scored ability under ppm-zero policy, not raw PPM bleed-through.

Not blocking MLE — Alex said so explicitly twice.

Locked / agreed actions

Item	Decision
Hero outcome	Inverted-U survives — primary scientific object
Gibbs SELECT	Gate cleared (PD20)
Box QC	Keep (dash + min 11 games)
Minutes floor	Drop sub-20 for locked panel
ppm-zero	Contrast / investigation only — not production policy
Season window	OK to drop 2011–2012 (ESPN depth); primary 2013+
ρ*	Have bracket value on hero panel (~0); optional edge — not gating paper
Next fit	MLE → λ, γ, t*

MLE — Alex lock (repeated)

1. **Write down the likelihood** (Charles says he already has a functional draft — simplify if overbuilt).
 2. **Then** use a solver (scipy / statsmodels / etc.) — only **three** parameters; should be smooth.
 3. **Do not** let H_sort forensics or ChatGPT code sprawl **block** this.
- Alex also pushed: code will need a **condense pass** after MLE — too much volume from AI-assisted scripts; expect a future sit-down to shrink to a succinct package for the paper.
- Related code**: `sports/scripts/pd21_draft_bernoulli_mle.py` (Bernoulli-softmax sketch — confirm matches Alex PD21 board factorization and K-draw story).
- Explainer stack**: `../3-Master_Plan/Alex_stuff/PD20_softmax_K_winners_explainer.md` · `../3-Master_Plan/Alex_stuff/PD20_K_draws_and_rho_explainer.md`

Army parallel track

Alex wants **anonymized Army export** so MBB work is not blocked forever — swap talent proxy (e.g. top-block ratio vs PPM) and reproduce the arc for the three-setting paper. Charles ~2–3 days on Army access last week; export “completely anonymized crap” for local + paper sharing when access returns (~1 month horizon mentioned).

Not blocking MBB MLE now.

Charles homework

- ☐ **Finish H_sort re-calibration run** (in progress Aug 18) — hero drop + ppm0lt20 on **current box-QC pipeline**; refresh AUTO slides if targets move
- ☐ **Write / simplify likelihood** for λ, γ, t (Gibbs/Bernoulli form Alex expects)
- ☐ **Run MLE** — grid + BFGS; season batches on locked drop panel
- ☐ **Lock written policy one-pager**: drop sub-20 (not ppm-zero); 2013+ window; slide 14 vs 15 roles
- ☐ **Optional ≤1 day**: H_sort numerator diagnostics + ppm-zero std-dev plot (Alex curiosity, not gate)
- ☐ **Optional**: `PD23_notes` → mirror under `3-Master_Plan/re_entry/` when stable
- ☐ **Army**: export script for anonymized panel when back on Rivanna/AWS

One-liners for next Alex touchpoint

Where we are: “Hero and Gibbs gate cleared; box QC and drop-at-20 locked; ρ* ≈ 0 on hero panel is modest H_sort fit — we’re fitting λ, γ, and t by MLE next.”

ppm-zero slide: “Slide 15 is contrast on the ppm0lt20 estimand — strong ρ* there shows the bracket machinery works when sorting targets are high; slide 14 is the locked hero calibration.”

If ρ comes up: “ρ is ASSIGN homophily, not a congestion dial; we can keep ρ at the bracket value or treat it as edge — hero doesn’t require ρ > 0.”

Artifacts

Item	Path
Transcript	transcripts/20260918_Paper_directions_23_otter_ai_transcript.docx
Campaign narrative	3-Master_Plan/re_entry/HEROs_and_PASSes/PD20_22_campaign_big_picture.md
Memo deck (read-aloud)	3-Master_Plan/re_entry/HEROs_and_PASSes/slides/CHAR_PD20_22_takeaways_memo_HAND.pptx
Hero p AUTO	slides/auto/CHAR_PD21_rho_hsort_calibrate_AUTO.pptx
ppm0 contrast AUTO	slides/auto/CHAR_PD21_rho_hsort_calibrate_ppm0lt20_AUTO.pptx
PD22 investigation	3-Master_Plan/re_entry/HEROs_and_PASSes/pd22_minutes/PD22_minutes_panel_investigation_todo.md